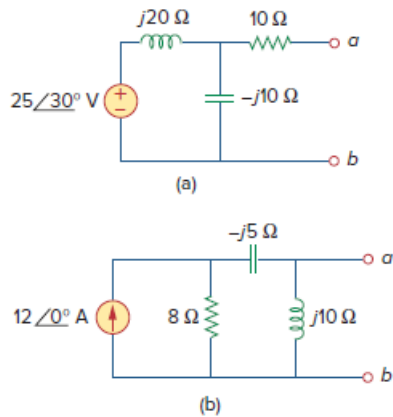


Problem

Find the Thevenin and Norton equivalent circuits at terminals a - b for each of the circuits in Fig. 10.98.

Figure 10.98:



Step-by-step solution

Step 1 of 12

(a)

Refer to Figure 10.98 (a) in the textbook.

Consider the following circuit to calculate the thevenin impedance:

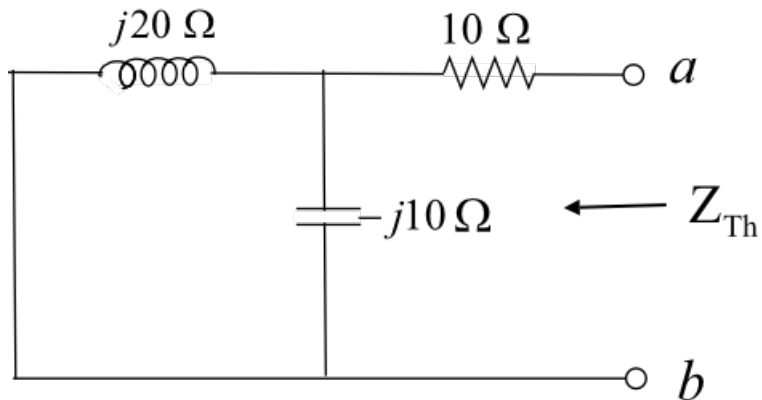


Figure 1

Step 2 of 12

From Figure 1, the thevenin impedance is,

$$\begin{aligned}
 Z_{th} &= (j20) \parallel (-j10) + 10 \\
 &= \frac{(j20)(-j10)}{j20 - j10} + 10 \\
 &= \frac{200}{j10} + 10 \\
 &= (10 - j20) \, \Omega
 \end{aligned}$$

Step 3 of 12

Consider the following circuit to calculate the thevenin voltage:

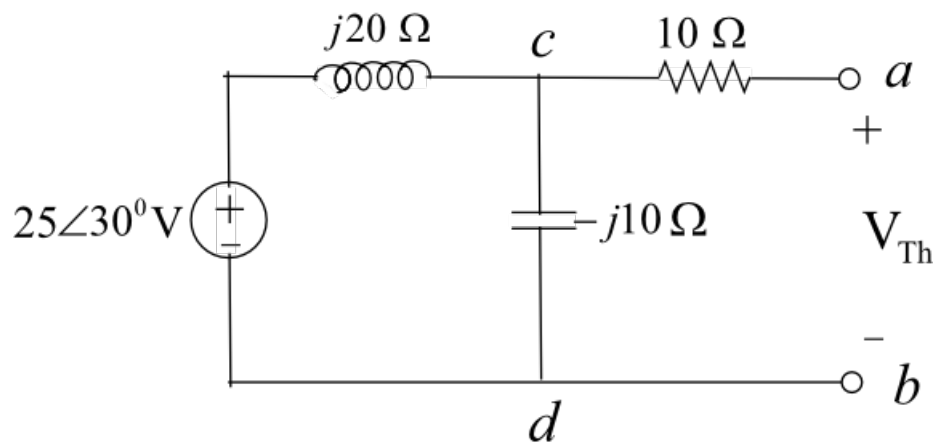


Figure 2

Step 4 of 12

Calculate the thevenin voltage using voltage division rule.

$$\begin{aligned}
 V_{th} &= (25\angle 30^\circ) \left(\frac{-j10}{j20 - j10} \right) \\
 &= (25\angle 30^\circ) \left(\frac{-j10}{j10} \right) \\
 &= -25\angle 30^\circ \\
 &= 25\angle -150^\circ \text{ V}
 \end{aligned}$$

Step 5 of 12

The Thevenin equivalent circuit is shown in Figure 3.

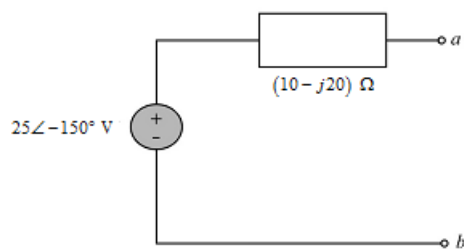


Figure 3

Step 6 of 12

Calculate the Norton equivalent current.

$$\begin{aligned} I_N &= \frac{V_{th}}{Z_{th}} \\ &= \frac{25\angle -150^\circ}{10 - j20} \\ &= 1.118\angle -86.56^\circ \text{ A} \end{aligned}$$

The Norton's equivalent circuit is shown in Figure 4:

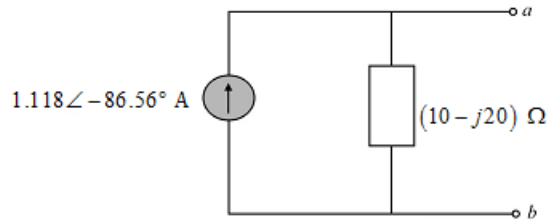


Figure 4

Step 7 of 12

(b)

Refer to Figure 10.98(b) in the textbook.

Consider the following circuit to calculate the thevenin impedance:

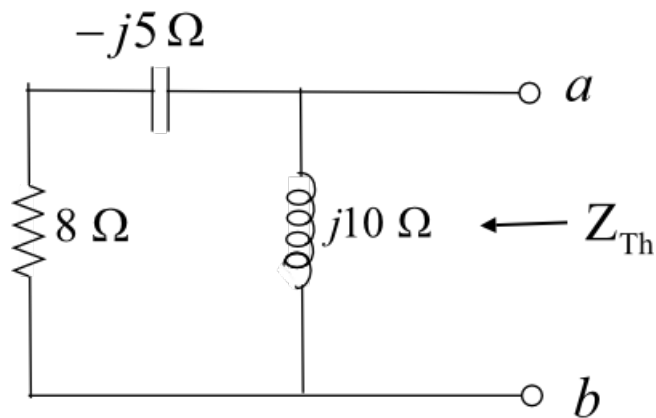


Figure 5

Step 8 of 12

From Figure 4, the thevenin impedance is,

$$\begin{aligned} Z_{Th} &= (8 - j5) \parallel j10 \\ &= \frac{(8 - j5)(j10)}{8 - j5 + j10} \\ &= \frac{50 + j80}{8 + j5} \\ &= 9 + j4.38 \Omega \end{aligned}$$

Step 9 of 12

Consider the following circuit to calculate the thevenin voltage:

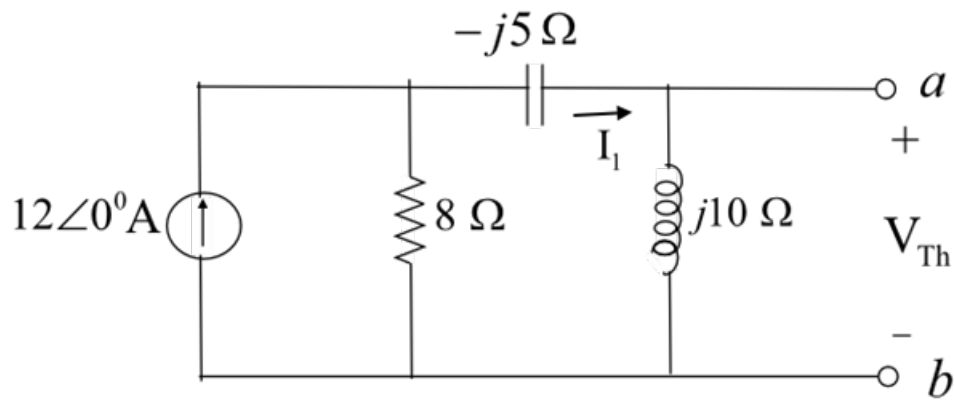


Figure 6

Step 10 of 12

From Figure 6, the current I_1 is,

$$\begin{aligned} I_1 &= (12\angle 0^\circ) \left(\frac{8}{8 - j5 + j10} \right) \\ &= (12\angle 0^\circ) \left(\frac{8}{8 + j5} \right) \\ &= 10.2\angle -32^\circ \text{ A} \end{aligned}$$

Therefore, the thevenin voltage is,

$$\begin{aligned} V_{th} &= I_1 (j10) \\ &= (10.2\angle -32^\circ) (10\angle 90^\circ) \\ &= 102\angle 58^\circ \text{ V} \end{aligned}$$

Step 11 of 12

The Thevenin equivalent circuit is shown in Figure 7.

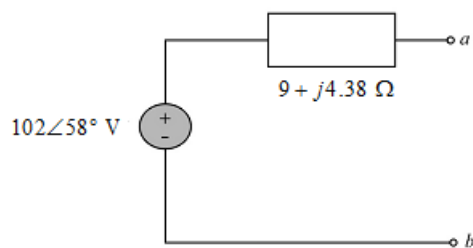


Figure 7

Step 12 of 12

Calculate the Norton equivalent current.

$$\begin{aligned} I_N &= \frac{V_{th}}{Z_{th}} \\ &= \frac{102\angle 58^\circ}{9 + j4.38} \\ &= 10.19\angle 32^\circ \text{ A} \end{aligned}$$

The Norton's equivalent circuit is shown in Figure 8:

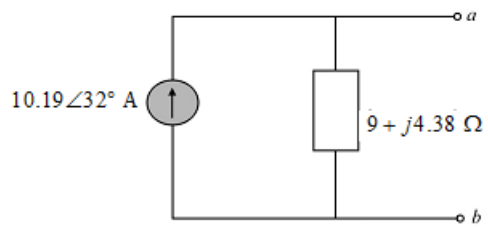


Figure 8